

Reliable Chest X-ray Diagnosis with Out-of-Distribution Detection using Deep Learning

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Abstract

Deep learning-based medical imaging systems have achieved high diagnostic performance but often fail under distribution shifts, producing high-confidence incorrect predictions. This project presents a reliable chest X-ray diagnosis system using DenseNet121 integrated with Out-of-Distribution (OOD) detection techniques. The model is trained on the NIH Chest X-ray dataset and evaluated on the CheXpert dataset to simulate distribution shift. Multiple OOD detection methods, including Maximum Softmax Probability (MSP), Energy-based scoring, and Mahalanobis distance, are evaluated. Risk mitigation and drift monitoring mechanisms are further incorporated to enhance safety.

1 Introduction

Deep learning has significantly transformed the field of medical imaging, enabling automated diagnosis with performance comparable to human experts in several tasks. In particular, convolutional neural networks (CNNs) have demonstrated strong capabilities in analyzing chest X-ray images for detecting multiple thoracic diseases such as pneumonia, cardiomegaly, and effusion.

Despite these advancements, a major limitation of deep learning systems is their inherent assumption that the training data and deployment data are drawn from the same underlying distribution. In real-world clinical environments, this assumption rarely holds true. Variations in imaging devices, acquisition protocols, patient demographics, and environmental conditions introduce significant distribution shifts between training and real-world data.

Such distribution shifts lead to a critical issue: models may produce highly confident but incorrect predictions when exposed to unfamiliar or unseen data. This is particularly dangerous in healthcare applications, where incorrect predictions can directly impact patient outcomes and clinical decisions.

In practical scenarios, the following factors contribute to distribution shift:

- Variability in imaging devices and hospital equipment

- Differences in patient populations across regions
- Presence of noise, blur, and contrast variations
- Changes in acquisition protocols and preprocessing pipelines

To ensure reliability and safety in deployment, it is essential to equip deep learning models with mechanisms that can detect and handle such distribution shifts. This leads to the concept of **Out-of-Distribution (OOD) detection**, which aims to identify inputs that do not belong to the training data distribution.

In this project, we propose a comprehensive framework for reliable chest X-ray diagnosis by integrating multiple components:

- A DenseNet121-based multi-label classification model for disease prediction
- Out-of-Distribution detection techniques, including Maximum Softmax Probability (MSP), Energy-based scoring, and Mahalanobis distance
- A risk mitigation mechanism that suppresses low-confidence predictions to reduce unsafe outputs
- A drift monitoring system that evaluates model robustness under simulated real-world corruptions such as noise, blur, and contrast changes

The key contribution of this work lies in combining classification, uncertainty estimation, and reliability mechanisms into a unified system. Unlike traditional approaches that focus solely on prediction accuracy, our framework emphasizes **trustworthiness, robustness, and safety**, which are critical requirements for deploying AI systems in real-world healthcare settings.

2 Literature Review

Deep learning has significantly advanced medical image analysis, particularly in chest X-ray diagnosis. Large-scale datasets such as ChestX-ray8 [6] and PadChest [1] have enabled the development of robust multi-label classification models. ChestX-ray8 introduced over 100,000 images with disease labels, achieving AUROC values ranging from 0.63 to 0.85 across different diseases. PadChest further improved dataset diversity by providing over 160,000 images with detailed annotations and radiology reports.

Convolutional Neural Networks (CNNs), particularly DenseNet [2], have demonstrated strong performance in image classification tasks. DenseNet achieves competitive ImageNet performance with Top-1 error around 25%, while maintaining efficient parameter usage. Building upon this, CheXNet [5] achieved an F1 score of 0.435 for pneumonia detection, surpassing average radiologist performance.

Despite high classification accuracy, these models assume that training and test data follow the same distribution. This assumption does not hold in real-world scenarios.

To address this, Liang et al. [3] proposed the Maximum Softmax Probability (MSP) method, achieving AUROC values in the range of 0.75–0.85, though with relatively high FPR@95TPR values (up to 60%). Liu et al. [4] introduced energy-based OOD detection, which significantly improves performance, achieving AUROC values above 0.90 and lower false positive rates.

Mahalanobis distance-based methods further attempt to model feature distributions, but their performance depends heavily on data representation and statistical assumptions.

Recent work by Zamzmi et al. [7] highlights that clinical AI systems degrade under real-world perturbations such as noise, blur, and contrast changes, emphasizing the importance of drift monitoring.

However, most existing works focus on classification or OOD detection independently. Our work differs by integrating:

- DenseNet-based classification
- Multiple OOD detection methods (MSP, Energy, Mahalanobis)
- Risk mitigation through confidence suppression
- Drift monitoring under simulated corruptions

This unified framework provides a more reliable and practical solution for real-world medical AI deployment.

3 Proposed Methodology

This section describes the complete pipeline of the proposed system, which integrates multi-label chest X-ray classification with Out-of-Distribution (OOD) detection, risk mitigation, and drift monitoring. The overall workflow consists of data preprocessing, model training, OOD detection, and reliability enhancement mechanisms.

3.1 Model Architecture

We employ DenseNet121 [2] as the backbone architecture for multi-label chest X-ray classification. DenseNet introduces dense connectivity between layers, where each layer receives feature maps from all preceding layers. This design improves feature reuse and enables efficient gradient propagation.

Formally, the output of the l^{th} layer is defined as:

$$x_l = H_l([x_0, x_1, x_2, \dots, x_{l-1}]) \quad (1)$$

where $[\cdot]$ denotes concatenation and $H_l(\cdot)$ represents a composite function consisting of Batch Normalization (BN), ReLU activation, and convolution.

3.1.1 Network Structure

DenseNet121 consists of:

- Initial convolution and pooling layer
- Four dense blocks with layers [6, 12, 24, 16]
- Transition layers between dense blocks
- Global average pooling layer
- Fully connected classification layer

Dense Blocks: Each dense block contains multiple convolutional layers, where each layer receives input from all previous layers. This results in feature reuse and reduces redundancy.

Growth Rate: Each layer produces k feature maps (growth rate), which are concatenated with previous features.

Transition Layers: Transition layers are placed between dense blocks and consist of:

- 1×1 convolution (for dimensionality reduction)
- 2×2 average pooling (for downsampling)

3.1.2 Adaptation for Multi-label Classification

The pretrained DenseNet121 is modified by replacing the final fully connected layer with a layer of size 15, corresponding to the number of disease classes.

Given an input image x , the network produces logits:

$$f(x) = [f_1(x), f_2(x), \dots, f_{15}(x)] \quad (2)$$

A sigmoid activation is applied to obtain class-wise probabilities:

$$p_i = \sigma(f_i(x)) = \frac{1}{1 + e^{-f_i(x)}} \quad (3)$$

The model is trained using Binary Cross Entropy (BCE) loss:

$$\mathcal{L} = - \sum_{i=1}^{15} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (4)$$

Feature Extraction for OOD Detection: Intermediate feature representations (typically from the penultimate layer) are extracted and used for OOD detection methods such as Mahalanobis distance.

3.2 Dataset

The model is trained on the NIH Chest X-ray dataset [6], which contains:

- 112,120 images from 30,805 patients
- 15 disease labels (multi-label setting)

Due to computational constraints, a subset of 24,999 images is used. The dataset is divided as follows:

- Training set: 16,403 images
- Validation set: 4,528 images
- Test set: 4,068 images

To evaluate robustness under distribution shift, the CheXpert dataset [5] is used as an Out-of-Distribution (OOD) dataset.

3.3 Data Preprocessing

Data preprocessing is a crucial step in preparing raw chest X-ray images for effective training of the DenseNet121 model. The original dataset consists of high-resolution medical images (typically 1024×1024), which require transformation to ensure compatibility with deep learning architectures and efficient computation.

The following preprocessing steps are applied:

- **Multi-label Binarization:** Each chest X-ray image can contain multiple disease labels simultaneously. Therefore, the labels are converted into a binary vector of size 15, where each position represents the presence (1) or absence (0) of a specific disease. This enables the model to perform multi-label classification instead of single-label prediction.
- **Image Resizing:** All input images are resized from their original high resolution (e.g., 1024×1024) to 224×224 pixels. This resizing is essential for several reasons:
 - DenseNet121 is pretrained on ImageNet, which expects input images of size 224×224
 - Reduces computational complexity and memory usage, allowing faster training and inference
 - Ensures uniform input dimensions across the dataset

The resizing operation preserves the overall anatomical structure of the chest X-ray while slightly reducing fine-grained details. However, this trade-off is acceptable as it enables efficient feature extraction and model compatibility.

- **Normalization:** Pixel values are normalized using ImageNet statistics (mean and standard deviation). This step ensures that the input distribution matches the distribution on which the pretrained DenseNet model was originally trained.

Normalization helps in:

- Stabilizing training
- Improving convergence speed
- Preventing large gradient updates

3.3.1 Preprocessing Impact

The transformation pipeline converts raw chest X-ray images into standardized inputs suitable for deep learning. The processed images maintain essential anatomical features while being optimized for model performance.

Overall, these preprocessing steps ensure that the input data is consistent, computationally efficient, and aligned with the requirements of the pretrained DenseNet architecture.

3.4 Class Imbalance Handling

The dataset exhibits significant class imbalance, where certain diseases have very few samples compared to others. To address this issue, class weights are computed based on inverse class frequency.

Let w_i denote the weight for class i :

$$w_i = \frac{1}{\text{frequency}(i)} \quad (5)$$

These weights are incorporated into the loss function to ensure that rare classes contribute more significantly during training.

3.5 Out-of-Distribution Detection

To improve model reliability, multiple Out-of-Distribution (OOD) detection techniques are applied. These methods estimate the confidence of model predictions and help identify inputs that deviate from the training distribution.

1. Maximum Softmax Probability (MSP):

The MSP method uses the maximum predicted probability as a confidence score:

$$\text{MSP}(x) = \max_k p(y = k|x) \quad (6)$$

Lower confidence values indicate a higher likelihood of the sample being OOD. While simple and computationally efficient, MSP often struggles to distinguish between uncertain in-distribution samples and true OOD inputs.

2. Energy-based Score:

Energy-based OOD detection uses the log-sum-exp of logits:

$$E(x) = -\log \sum_k e^{f_k(x)} \quad (7)$$

This approach leverages raw logits instead of probabilities, providing better-calibrated confidence estimates and improved separation between in-distribution and OOD samples.

3. Mahalanobis-based OOD Detection:

Mahalanobis distance measures how far a sample lies from the distribution of training data in feature space. Given a feature representation $f(x)$, the Mahalanobis distance for class k is defined as:

$$D_k(x) = (f(x) - \mu_k)^T \Sigma^{-1} (f(x) - \mu_k) \quad (8)$$

Here:

- μ_k represents the **mean feature vector** (centroid) of class k , computed from training data.
- Σ represents the **covariance matrix** of the feature representations, capturing the spread and correlation of features.

The covariance matrix Σ encodes how different feature dimensions vary with respect to each other, and its inverse Σ^{-1} scales the distance based on feature importance and correlation.

The final OOD score is computed as:

$$\text{Score}(x) = -\min_k D_k(x) \quad (9)$$

This score represents the negative distance to the closest class distribution. A higher score indicates that the sample is closer to the training distribution (in-distribution), while a lower score suggests that the sample is likely out-of-distribution.

To improve robustness, multiple variants of Mahalanobis distance are explored:

- **PCA (128 dimensions, single Gaussian):** Features are projected into a lower-dimensional space and modeled using a single global Gaussian distribution.
- **PCA (256 dimensions, conditional Gaussian):** Class-conditional Gaussian distributions are estimated to better capture intra-class variations.
- **1024-dimensional Ledoit-Wolf (global):** High-dimensional features are used with shrinkage-based covariance estimation for numerical stability.
- **1024-dimensional Ledoit-Wolf (multi-label):** Separate Gaussian distributions are estimated for each label in the multi-label setting, enabling finer modeling of disease-specific features.

Mahalanobis-based methods depend heavily on feature representation quality and covariance estimation accuracy. While they provide valuable insights into feature-space structure, they often underperform compared to energy-based methods due to stronger distributional assumptions.

Summary: All three methods (MSP, Energy, and Mahalanobis) are used to compute confidence scores. These scores are further utilized for OOD detection, risk mitigation, and drift analysis in the proposed system..

3.6 Risk Mitigation Strategy

To reduce unsafe predictions, a threshold-based rejection mechanism is introduced. Let τ be a predefined confidence threshold:

- If confidence $\geq \tau$, prediction is accepted
- If confidence $< \tau$, prediction is rejected

Rejected samples are flagged for further human evaluation, ensuring safer deployment in clinical settings.

3.7 Drift Monitoring

To simulate real-world conditions, data corruption is introduced using:

- Gaussian noise
- Contrast variation
- Blur effects

The model is evaluated under increasing levels of corruption severity. Drift monitoring analyzes:

- Change in prediction confidence
- Increase in OOD detection rate

This allows assessment of model robustness under distribution shifts.

3.8 Overall Pipeline

The complete workflow of the proposed system is as follows:

1. Input chest X-ray image
2. Preprocessing and normalization
3. DenseNet-based feature extraction and classification
4. OOD score computation (MSP, Energy, Mahalanobis)
5. Confidence-based filtering (risk mitigation)
6. Drift monitoring for robustness evaluation

This integrated pipeline ensures not only high classification performance but also improved reliability and safety under real-world deployment conditions.

4 Experimental Results

4.1 DenseNet Classification Performance

4.1.1 Performance on NIH Test Set (In-Distribution)

Observation: The model achieves strong performance across multiple classes, with highest AUROC observed for Hernia (0.9035) and Cardiomegaly (0.8801). Lower performance on Pneumonia and Infiltration indicates class imbalance and difficulty in feature learning for rare or complex diseases.

Label	AUROC	Positives
Atelectasis	0.7024	465
Cardiomegaly	0.8801	378
Effusion	0.7623	587
Infiltration	0.6539	1061
Mass	0.7619	148
Nodule	0.6903	207
Pneumonia	0.6397	93
Pneumothorax	0.8122	303
Consolidation	0.6502	262
Edema	0.7658	121
Emphysema	0.8501	120
Fibrosis	0.7601	139
Pleural_Thickening	0.7568	202
Hernia	0.9035	26
Macro AUROC	0.7564	

Table 1: DenseNet121 Performance on NIH Test Set

Label	AUROC	Positives
Cardiomegaly	0.6524	66
Edema	0.7708	42
Consolidation	0.8369	32
Pneumonia	0.5110	8
Atelectasis	0.8068	75
Pneumothorax	0.7136	7
Macro AUROC	0.7153	

Table 2: DenseNet121 Performance on CheXpert (OOD)

4.1.2 Performance on CheXpert Dataset (OOD)

Observation: A drop in performance from 0.8087 to 0.7153 macro AUROC confirms sensitivity to distribution shift. Significant degradation is observed in Pneumonia detection (0.5110), highlighting domain mismatch.

4.2 OOD Detection Performance (MSP and Energy)

Method	AUROC	FPR@95TPR
MSP	0.7547	0.5656
Energy Score	0.8061	0.3799

Table 3: OOD Detection using MSP and Energy-based Methods

Observation: Energy-based OOD detection outperforms MSP in both AUROC and

FPR@95TPR. Lower FPR indicates fewer false positives at high recall, making it more reliable for real-world deployment.

4.3 Mahalanobis-based OOD Detection

Method	AUROC	FPR
Mahalanobis (128 PCA, single Gaussian)	0.5869	0.8478
Mahalanobis (256 PCA, conditional)	0.6480	0.7928
Mahalanobis (1024 dims, Ledoit-Wolf, global)	0.7683	0.5825
Mahalanobis (1024 dims, Ledoit-Wolf, multi-label)	0.7520	0.5922

Table 4: Mahalanobis-based OOD Detection Variants

Observation: Among Mahalanobis variants, the Ledoit-Wolf global covariance approach performs best (AUROC = 0.7683). However, all Mahalanobis methods underperform compared to energy-based detection, indicating limitations in Gaussian assumptions for high-dimensional feature distributions.

4.4 Overall Discussion

The experimental results highlight the following key insights:

- DenseNet121 achieves strong multi-label classification performance on in-distribution data.
- Performance degradation on CheXpert confirms the impact of distribution shift.
- Energy-based OOD detection provides the best trade-off between AUROC and FPR.
- Mahalanobis-based methods are sensitive to feature distribution assumptions and show comparatively lower performance.

5 Risk Mitigation

Deep learning models often produce high-confidence predictions even for incorrect or out-of-distribution inputs. In safety-critical domains such as medical diagnosis, such errors can lead to severe consequences. Therefore, it is essential to incorporate mechanisms that reduce the likelihood of high-confidence incorrect predictions.

5.1 Confidence-based Suppression

In this work, risk mitigation is implemented using a threshold-based rejection mechanism based on confidence scores derived from energy-based OOD detection.

Let τ denote a predefined confidence threshold. The decision rule is defined as:

- If confidence $\geq \tau$, the prediction is accepted
- If confidence $< \tau$, the prediction is suppressed

Suppressed predictions are flagged for further human evaluation. This ensures that uncertain or potentially incorrect predictions are not directly used in decision-making, thereby improving system reliability.

5.2 Energy-based Risk Control

Energy scores provide a more reliable estimate of model confidence compared to softmax probabilities. By adjusting the suppression threshold, we can control the trade-off between:

- **Coverage:** Fraction of inputs for which predictions are made
- **Risk:** Probability of incorrect predictions

5.3 Experimental Analysis

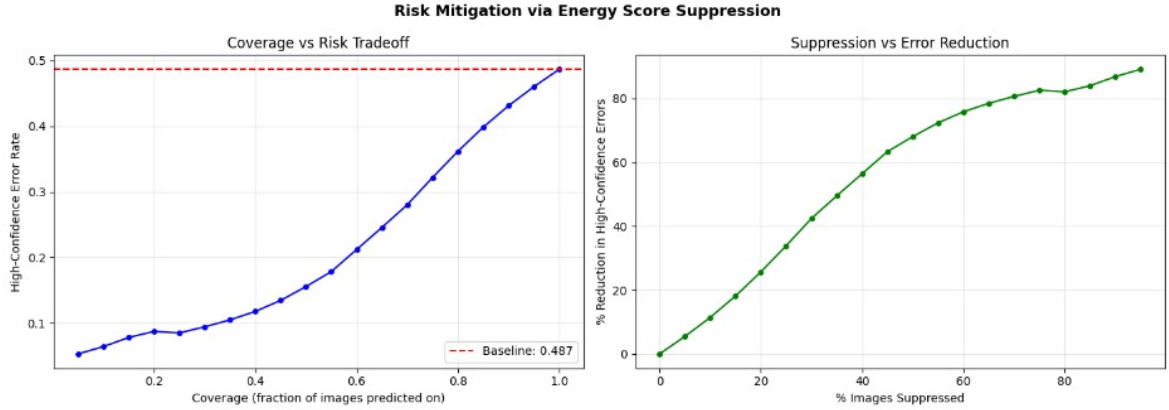


Figure 1: Risk Mitigation via Energy Score Suppression: (Left) Coverage vs Risk Trade-off, (Right) Suppression vs Error Reduction

The left plot shows the relationship between coverage and high-confidence error rate. As coverage increases (i.e., fewer samples are suppressed), the error rate also increases. This demonstrates the trade-off between making more predictions and maintaining reliability.

The right plot illustrates the percentage reduction in high-confidence errors as a function of the percentage of suppressed samples. As more uncertain samples are suppressed, the reduction in errors increases significantly, reaching up to approximately 90%.

5.4 Discussion

These results highlight that:

- Energy-based suppression effectively reduces high-confidence incorrect predictions

- A small reduction in coverage can lead to a substantial decrease in error rate
- The system can be tuned based on application requirements (e.g., higher safety vs higher coverage)

Overall, risk mitigation plays a crucial role in making the model safer and more reliable for real-world deployment, particularly in clinical settings where incorrect predictions must be minimized.

6 Drift Monitoring

6.1 Motivation

In real-world clinical environments, data distributions often change over time due to variations in imaging devices, acquisition protocols, and patient conditions. This phenomenon, known as **data drift**, can significantly degrade model performance and reliability.

To ensure safe deployment, it is essential to continuously monitor how model behavior changes under such distribution shifts.

6.2 Simulation of Distribution Shift

To simulate real-world drift, controlled perturbations are applied to input images:

- **Gaussian Noise:** simulates sensor noise and acquisition artifacts
- **Contrast Reduction:** simulates poor imaging conditions
- **Blur:** simulates motion blur or defocus effects

Each corruption type is applied with increasing **severity levels** ranging from 0 to 1.

6.3 Evaluation Metric

Drift is quantified using the **OOD detection rate**, which measures the fraction of samples identified as out-of-distribution. A threshold of 20% is used as an alert level to indicate significant distribution shift.

6.4 Experimental Results

6.5 Analysis

The results show a clear trend across all corruption types:

- As corruption severity increases, the OOD detection rate increases consistently
- The curve crosses the alert threshold (20%) at moderate severity levels

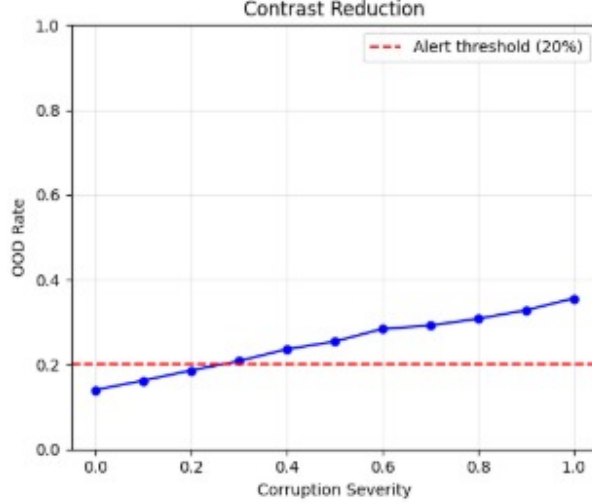


Figure 2: Contrast Reduction

Contrast Reduction: A gradual increase in OOD rate is observed, showing that the model is moderately robust to contrast variations but still affected at higher severity levels.

The OOD rate increases steadily and sharply at higher severity levels, highlighting sensitivity to noise and pixel-level perturbations for both the cases of Blur as well as Gaussian noise as well.

Overall, drift monitoring provides an essential safety mechanism by continuously evaluating model performance under varying conditions, ensuring that unreliable predictions can be detected and handled appropriately in real-world deployment.

7 Summary

This section presents a consolidated overview of the experimental findings, system behavior, and future directions of the proposed medical AI framework.

7.1 Overall System Performance

The DenseNet121 model demonstrates strong performance on the NIH test dataset, achieving a macro AUROC of **0.7564**. High AUROC values for diseases such as Hernia (0.9035) and Cardiomegaly (0.8801) indicate that the model effectively learns discriminative features for well-represented classes. However, comparatively lower performance on Pneumonia and Mass highlights challenges related to class imbalance and subtle inter-class feature similarities.

When evaluated on the CheXpert dataset, the macro AUROC drops to **0.7153**, clearly demonstrating the effect of distribution shift. The significant degradation in Pneumonia detection (AUROC = 0.5110) further emphasizes the difficulty of generalizing to unseen data distributions and reinforces the need for robustness-focused techniques.

7.2 Effectiveness of OOD Detection

Among the evaluated Out-of-Distribution (OOD) detection methods, the energy-based approach consistently outperforms MSP and Mahalanobis methods. Its superior performance in both AUROC and FPR@95TPR can be attributed to the use of logits, which provide better-calibrated confidence estimates compared to softmax probabilities.

The MSP method, although simple and computationally efficient, suffers from over-confidence issues, making it less reliable for distinguishing uncertain in-distribution samples from true OOD inputs.

Mahalanobis-based approaches, including PCA-based and Ledoit-Wolf variants, show comparatively lower performance. While these methods aim to model feature distributions in latent space, their effectiveness is limited by assumptions of Gaussianity and sensitivity to feature representation quality.

7.3 Impact of Risk Mitigation

The introduction of a risk mitigation strategy using energy-based suppression significantly improves system safety. By rejecting low-confidence predictions, the model effectively reduces high-confidence incorrect outputs, which are critical to avoid in medical applications.

Experimental results demonstrate that even a small reduction in coverage leads to a substantial decrease in error rates. This highlights an important trade-off: prioritizing reliability over coverage is essential in safety-critical domains such as healthcare.

7.4 Drift Monitoring Insights

Drift monitoring experiments reveal that the model is sensitive to real-world perturbations such as Gaussian noise, blur, and contrast variations. As corruption severity increases, the OOD detection rate increases consistently across all scenarios.

The observation that OOD rates exceed the alert threshold at moderate severity levels indicates that even minor distribution shifts can significantly affect model reliability. This underscores the importance of continuous monitoring in deployed systems to detect performance degradation early.

7.5 Integrated System Perspective

The proposed system combines multiple components to address different challenges:

- DenseNet121 provides strong baseline performance for multi-label classification
- OOD detection identifies inputs that deviate from the training distribution
- Risk mitigation reduces unsafe high-confidence predictions
- Drift monitoring enables detection of distribution shifts over time

Together, these components form a comprehensive framework that moves beyond traditional accuracy-focused approaches and emphasizes robustness, safety, and trustworthiness.

7.6 Limitations

Despite the promising results, certain limitations remain:

- Performance degrades under significant distribution shifts
- Mahalanobis-based methods are sensitive to feature distribution assumptions
- Class imbalance continues to affect performance for rare diseases

7.7 Future Work

Future improvements can focus on:

- Enhancing Mahalanobis-based methods with better feature representations
- Incorporating calibration techniques to improve confidence estimation
- Deploying the system in real clinical environments for validation
- Exploring advanced architectures such as transformer-based models

7.8 Conclusion

In conclusion, this work presents a reliable and robust medical AI system that integrates classification, OOD detection, risk mitigation, and drift monitoring into a unified framework. The results demonstrate that incorporating reliability-focused mechanisms significantly enhances model safety and practical usability. This approach represents an important step toward deploying trustworthy deep learning systems in real-world healthcare applications.

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